

BMS Reconfigurable Project — Shopping List (Corrected)

Updated: Incorporates all corrections from M0 findings, GPIO map revision, and component changes. Prices in INR, estimates only — verify before ordering.

⚠ Corrections Applied vs Original BOM

Item	Original BOM	Corrected	Why
IRF4905 qty	6 + 2 spare = 8	8 needed, 0 spare	Q4 (load disconnect) added from M0 — needs 2 more
2N2222A qty	3 + 2 spare = 5	4 needed, 1 spare = 5	Q4 needs its own level shifter
Current sensor	INA226 module	INA260 module	Already in hand; built-in 2 mΩ shunt, no replacement needed
Heatsink qty	3	8 needed (one per MOSFET) + 2 spare = 10	8 MOSFETs in circuit, all need heatsinks at 10A peak
100kΩ resistor qty	3 + 2 spare = 5	4 needed + 2 spare = 6	Q4 base pull-down added
1kΩ resistor qty	3 + spare	4 needed + spare = buy 10	Q4 base resistor added
10kΩ resistor qty	15 + spare = 20	16 needed + spare = buy 20	Q4 gate pull-up added (4 pull-ups + 6 NTC + 6 divider = 16)
Bulk capacitor	Discrepancy: plan says 1000 μF, checklist says 4700 μF	4700 μF	Sag calc: at 1000 μF, 5A × 2ms = 10V sag (kills the bus). At 4700 μF = 2.1V sag (safe)
BTS7960 qty	1	2	4WD differential steering needs 2 modules

PRIORITY 1 — Order First (Needed for M1: Switching + Sensing)

These are on the critical path. Order on Day 1.

#	Component	Specification	Qty	Est. ₹/ unit	Est. Total	Source
1	STM32 Nucleo board	NUCLEO-F446RE (STM32F446RET6)	1	1,600	1,600	Mouser India
2	P-MOSFET	IRF4905 (P-ch, TO-220, -55V, -74A)	10	50	500	Amazon IN / Robu
3	NPN transistor	2N2222A (TO-92)	5	5	25	Local / Amazon
4	TO-220 heatsink	Clip-on aluminium (~20°C/W)	10	15	150	Amazon IN
5	R — 1kΩ 1%	Metal film 1/4W (NPN base resistor)	10	1	10	Local
6	R — 10kΩ 1%	Metal film 1/4W (gate pull-up + NTC pull-up + divider bottom)	20	1	20	Local
7	R — 33kΩ 1%	Metal film 1/4W (voltage divider top)	10	1	10	Local
8	R — 100kΩ 1%	Metal film 1/4W (NPN base pull-down — SAFETY CRITICAL)	6	1	6	Local
9	Precharge resistor	10Ω 2W wire-wound	2	8	16	Local
10	Bulk capacitor	4700 μF 25V electrolytic, radial	2	40	80	Local / Amazon
11	NTC thermistor	10K bead type with wire leads	8	15	120	Amazon IN
12	Breadboard	830-point (for level shifter testing before soldering)	1	80	80	Local / Amazon
13	Zero PCB	10cm × 15cm perfboard (minimum)	3	40	120	Local

Note on IRF4905 qty: You need exactly 8 in circuit (4 back-to-back pairs for Q1-Q4). Buying 10 gives you 2 spares. Handle with ESD precautions — a dead MOSFET means waiting for reorder.

Note on INA260: You already have these. Verify you have **2 modules** with different I2C addresses (check A0/A1 pin strapping on each module).

PRIORITY 2 — Order with Priority 1 or Shortly After

#	Component	Specification	Qty	Est. ₹/ unit	Est. Total	Source
14	Li-ion cells	Samsung INR18650-30Q (3Ah, 15A)	6	250	1,500	Amazon IN / Robu
15	Cell holders	Single 18650 holder with solder tabs	6	25	150	Amazon IN
16	Buck converter	LM2596 module (4.5-40V in, adj. out, 3A)	1	70	70	Amazon IN
17	LDO regulator	AMS1117-3.3 module (5V → 3.3V)	2	20	40	Local
18	Polyfuse	15A resettable PPTC	2	30	60	Amazon IN
19	XT60 connector	Male + female pair (kill switch loop)	2	40	80	Amazon IN
20	XT30 connector	Male + female pair (cell connections)	6	25	150	Amazon IN

PRIORITY 3 — Wiring and Consumables

#	Component	Specification	Qty	Est. ₹/ unit	Est. Total	Source
21	Power wire	14 AWG silicone, red	2m	15/m	30	Amazon IN
22	Power wire	14 AWG silicone, black	2m	15/m	30	Amazon IN
23	Signal wire	22 AWG, assorted colors	3m	8/m	24	Amazon IN
24	Header pins	Male + female breakaway strips	4 strips	15	60	Local
25	Consumables kit	Solder, flux, solder wick, heat shrink, Kapton tape, zip ties	1 set	200	200	Local
26	M3 standoffs	Assorted nylon/metal standoffs + screws	1 set	50	50	Local

PRIORITY 4 — Communication (Optional, order if time permits)

#	Component	Specification	Qty	Est. ₹/unit	Est. Total	Source
27	CAN transceiver	SN65HVD230 module	1	90	90	Amazon IN
28	OLED display	SSD1306 128×64 I2C	1	150	150	Amazon IN
29	120Ω resistor	CAN bus termination	2	1	2	Local

PRIORITY 5 — Rover Platform (Can order in parallel, not on M1 critical path)

#	Component	Specification	Qty	Est. ₹/unit	Est. Total	Source
30	4WD chassis	Kit with 4× DC gear motors	1	1,500	1,500	Amazon IN / Robu
31	Motor driver	BTS7960 module (43A)	2	400	800	Amazon IN
32	RC controller	FlySky FS-i6 transmitter + receiver	1	3,500	3,500	Amazon IN

 **Motor Decision Still Open:** TT plastic gearmotors (~0.25A each, ~1A total) won't stress-test the BMS. For a meaningful demo you want 12V metal gearmotors (~1-1.5A each, ~4-6A total). Confirm motor spec before ordering chassis kit — some kits include motors, some don't.

PRIORITY 6 — PCB Fabrication (After zero PCB prototype validates logic)

#	Component	Specification	Qty	Est. ₹/unit	Est. Total	Source
33	Custom PCB	2-layer, 2oz copper, JLCPCB (set of 5)	1 order	500	500	JLCPCB / PCBWay

Cost Summary

Category	Estimated Total (₹)
BMS Electronics (items 1-13)	~2,737
Cells + Power + Safety (items 14-20)	~2,050
Wiring + Consumables (items 21-26)	~394
Communication — optional (items 27-29)	~242
Rover Platform (items 30-32)	~5,800
PCB Fabrication — later (item 33)	~500
Grand Total (everything)	~₹11,723
BMS + Cells only (no rover, no PCB)	~₹5,181

Already In Hand (Do Not Re-Order)

☐ INA260 current sensor modules × 2 — verify both have unique I2C addresses

Pre-M1 Checklist (Before You Start Building)

- ☐ All items from Priority 1 physically in hand
- ☐ INA260 modules: confirm 2 with different I2C addresses
- ☐ IRF4905: count all 10, store in anti-static bag
- ☐ 2N2222A: count all 5, verify TO-92 package (pin order: E-B-C for Style 1, **verify on your batch**)
- ☐ Multimeter available (voltage + current verification at every step)
- ☐ Oscilloscope access confirmed (lab or personal — needed for switching transient check)
- ☐ Bench power supply available (to substitute for cells during M1)
- ☐ 10Ω 50W power resistor for bench load testing (borrow from lab if possible)

Resistor Usage Breakdown (For Verification)

Value	Per switch (×4)	Voltage dividers (×6)	NTC pull-ups (×6)	Total Needed	Buying
1kΩ 1%	4 (NPN base)	—	—	4	10
10kΩ 1%	4 (gate pull-up)	6 (divider bottom)	6 (NTC pull-up)	16	20
33kΩ 1%	—	6 (divider top)	—	6	10
100kΩ 1%	4 (base pull-down)	—	—	4	6